

Business Analytics Case Study

Understanding and Reducing Customer Churn at a Telecommunications Company

Dataset: Telco Customer Churn (7,043 customers, 21 variables)

AnalystLab Africa Data Analytics Internship Program — Week 5

Batch B · Business Analytics Case Study

1. Introduction

Customer retention is one of the most important drivers of profitability in the telecommunications industry, where acquiring a new customer typically costs far more than retaining an existing one. This report presents a business analytics case study built on a dataset of 7,043 customers from a telecommunications company, examining who churns, why they churn, and what the business can do about it.

The analysis combines exploratory data analysis, distribution and correlation analysis, and segment-level churn-rate comparisons to translate raw customer records into insights a business manager can act on. All figures below are drawn directly from the dataset provided.

2. Business Problem Statement

The company is losing a meaningful share of its customer base to churn. Understanding which customers are most likely to leave, and why, allows the business to prioritize retention spend where it will have the greatest impact rather than applying blanket offers across the whole base. This report addresses two questions:

- Why are customers leaving the company?
- Which factors contribute most to customer churn, and which customer segments should be prioritized for retention efforts?

3. Dataset Description

The dataset contains 7,043 customer records across 21 columns, covering demographics (gender, senior citizen status, partner, dependents), account information (tenure, contract type, payment method, billing), subscribed services (phone, internet, online security, tech support, streaming), and billing amounts (monthly and total charges), alongside the target variable, Churn (Yes/No).

Attribute	Detail
Rows (customers)	7,043
Columns (variables)	21
Missing values found	11 blank TotalCharges records (customers with 0 tenure); imputed with the column median
Duplicate rows	0
Target variable	Churn (Yes / No)
Overall churn rate	26.54% churned (1,869 customers); 73.46% retained (5,174 customers)

The dataset is moderately imbalanced (roughly 1 churned customer for every 2.8 retained), which is typical for churn data and was accounted for by analyzing churn as a percentage within each segment rather than raw counts.

4. Data Analysis Process

The analysis followed a structured process using Python (pandas, seaborn, matplotlib) in a Colab notebook:

- Data cleaning: converted TotalCharges to numeric, imputed 11 missing values with the median, and confirmed there were no duplicate records.
- Exploratory analysis: reviewed data types, summary statistics, and the distribution of churn, demographics, and account variables using bar charts, pie charts, and histograms.
- Segment analysis: cross-tabulated churn rate against every categorical variable (contract, internet service, payment method, add-on services, demographics) to isolate which segments churn most.

- Numerical comparison: compared average tenure, monthly charges, and total charges between churned and retained customers using box plots and grouped means.
- Correlation analysis: built a correlation heatmap across encoded numeric and binary variables to confirm which factors move together with churn.

5. Key Findings

5.1 Contract Type Is the Strongest Driver of Churn

Churn rate falls sharply as contract commitment increases:

Contract Type	Churn Rate
Month-to-month	42.71%
One year	11.27%
Two year	2.83%

Month-to-month customers are churning at roughly 15 times the rate of two-year contract customers, making contract type the single clearest predictor in the dataset.

5.2 Fiber Optic Internet Customers Churn Nearly 3x More Than DSL

Internet Service	Churn Rate
Fiber optic	41.89%
DSL	18.96%
No internet service	7.40%

Despite being the premium, higher-revenue offering, fiber optic customers churn at more than double the rate of DSL customers, and nearly 6 times the rate of customers with no internet service — suggesting dissatisfaction with price, reliability, or competitive alternatives in that segment.

5.3 Electronic Check Payers Churn Most

Payment Method	Churn Rate
Electronic check	45.29%
Mailed check	19.11%
Bank transfer (automatic)	16.71%
Credit card (automatic)	15.24%

Customers who pay by electronic check churn nearly 3 times more than those on automatic payment methods, pointing to a link between payment friction (or the type of customer who avoids autopay) and churn risk.

5.4 Tenure and Spend Patterns

Metric	Churned Customers	Retained Customers
Average tenure (months)	17.98	37.57

Average monthly charges (\$)	74.44	61.27
Average total charges (\$)	1,531.80	2,552.88

Customers who churn leave, on average, less than half as far into their tenure as those who stay, despite paying higher monthly charges. This is consistent with new customers on premium plans being the most flight-risk group, while long-tenured customers — even at lower monthly rates — are comparatively loyal.

5.5 Lack of Add-On Services Correlates with Higher Churn

Service	Churn Rate — No	Churn Rate — Yes
Online Security	41.8%	14.6%
Tech Support	41.6%	15.2%
Device Protection	39.1%	22.5%

Customers without online security or tech support churn at roughly 2.5-3x the rate of customers who have these add-ons, suggesting these services build stickiness rather than simply correlating with higher-value customers.

5.6 Demographic Patterns

- Senior citizens churn markedly more than non-seniors: 41.7% vs 23.6%.
- Customers without a partner churn more than those with one: 33.0% vs 19.7%.
- Customers without dependents churn more than those with dependents: 31.3% vs 15.5%.
- Gender shows almost no relationship with churn (26.9% female vs 26.2% male), and can be deprioritized as a targeting variable.

6. Business Insights

- Churn risk is concentrated, not evenly spread: The highest-risk profile is a new, month-to-month customer on fiber optic internet, paying by electronic check, without online security or tech support add-ons. This combination compounds across nearly every risk factor identified.
- Premium customers are not always loyal customers: Fiber optic and higher monthly-charge customers churn more, not less, which challenges the assumption that higher spend signals higher satisfaction — it may instead signal price sensitivity or unmet service expectations.
- Contract length is a retention lever, not just a sales outcome: The gap between month-to-month (42.71%) and two-year (2.83%) churn is too large to be coincidental; commitment structure itself appears to reduce churn, independent of the customer's underlying satisfaction.
- Add-on services function as retention tools: Online security and tech support are associated with substantially lower churn, suggesting these are not just revenue add-ons but also engagement and stickiness mechanisms.
- At-risk segments (seniors, single customers, customers without dependents) may have different service needs or price sensitivity, and a one-size-fits-all retention approach is unlikely to work equally well across all of them.

7. Recommendations

- **Incentivize longer contracts:** Offer targeted discounts or perks to month-to-month customers who upgrade to one- or two-year contracts, prioritizing customers in their first 12 months of tenure — the highest-risk window.
- **Investigate the fiber optic experience:** Given fiber optic customers churn at nearly 42%, conduct a focused review of pricing, reliability, and competitive positioning for this service line before assuming the issue is purely price.

- **Reduce friction around electronic check payments:** Migrate electronic check users toward automatic payment methods (bank transfer or credit card) through incentives, since these methods correlate with roughly 3x lower churn.
- **Bundle or promote online security and tech support:** Proactively offer these add-ons, especially to new and fiber optic customers, given their strong association with reduced churn.
- **Build a targeted early-tenure retention program:** Since churned customers leave at an average of just 18 months versus 38 months for retained customers, concentrate proactive outreach, check-ins, or loyalty incentives within a customer's first year.
- **Tailor outreach for at-risk demographics:** Design retention messaging and offers specifically for senior citizens and single customers without dependents, who show meaningfully higher churn than their counterparts.

8. Conclusion

This analysis shows that churn at the company is not random — it is concentrated among new, month-to-month, fiber optic customers who pay by electronic check and lack add-on services. Contract type, tenure, internet service type, payment method, and service add-ons are the clearest, most actionable drivers identified in the data. Acting on these findings — through contract incentives, a fiber optic service review, payment migration, add-on bundling, and an early-tenure retention program — offers the company a concrete, data-driven path to reducing its 26.54% churn rate and protecting recurring revenue.